Elementary Matrices

* **Lemma 1.1** Matrix multiplication is associative ie $A.(B.C) = (A.B).C$

* **Lemma 1.2** If $P = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$, $Q = \left[\begin{array}{c|c} A' & B' \\ \hline C' & D' \end{array}\right]$ are matrices s.t (i) $\exists R = P.Q$ and (ii) # columns of A = # rows of A'

  then $R = \begin{bmatrix} AA' + BC' & AB' + BD' \\ CA' + DC' & CB' + DD' \end{bmatrix}$

* **Defn 1.3** A matrix A is called invertible if $\exists\ B$ s.t $A.B = B.A = I$

* **Lemma 1.4** A is invertible $\iff$ A^T is invertible

* **Lemma 1.5** If A, B and C are matrices s.t $C = A.B$ and both A and B are invertible $\Longrightarrow$ C is invertible

* **Lemma 1.6** Elementary Matrices $\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} / \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix} / \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix}$

  → Effect of left and right multiplication

  → They are invertible

* **Defn 1.7** Reduced row echelon form (rref):

  ① The first non zero element in a row, ie the pivot, is 1. All the elements above and below a pivot are zero

  ② If row i and row j are two rows of rref s.t (i) $j > i$ and (ii) they both have pivot elements then row j pivot lies to the right of row i pivot

  ③ If row i is all zeros then all the rows below it are also all zeros.

* **Lemma 1.8**

  A (square matrix)

  ↙ invertible — non invertible ↘

  $\text{rref}(A) = I$; $\text{rref}(A) = \begin{bmatrix} P & Q \\ 0 & 0 \end{bmatrix}$ and also $\exists$ an invertible matrix M s.t $\text{rref}(A) = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} M$

* **Theorem 1.9** If A, B, C are square matrices s.t $C = A.B$

  C is invertible $\iff$ Both A and B are invertible

  Hint Use 1.8 and 1.5

* Lemma 1.10 — A is an invertible matrix $\iff$ A is product of elementary matrices

* Defn 1.11 — Determinant : $\mathbb{R}^{n\times n} \longrightarrow \mathbb{R}$ s.t

① $\det(I) = 1$

② $\det\left(\begin{bmatrix} \vdots \\ a\rho + b\mathcal{Q} \\ \vdots \end{bmatrix}\right) = a\det\left(\begin{bmatrix} \vdots \\ \rho \\ \vdots \end{bmatrix}\right) + b\det\left(\begin{bmatrix} \vdots \\ \mathcal{Q} \\ \vdots \end{bmatrix}\right)$, ie det is linear in rows

③ If two rows of A are same then $\det(A) = 0$

* Lemma 1.12 — If E is an elementary matrix and A are square matrix then

$$\det(EA) = \det(E)\det(A)$$

* Theorem 1.13 — If A and B are square matrices of same size then

$$\det(AB) = \det(A)\det(B)$$ Hint 1.12, 1.10

* Lemma 1.14 — If A is a square matrix then $\det(A) = \det(A^T)$

* Theorem 1.15 — If $A, \rho, \mathcal{Q}$ are square matrices s.t $A = \begin{bmatrix} \rho & 0 \\ 0 & \mathcal{Q} \end{bmatrix}$ then

$$\det(A) = \det(\rho)\det(\mathcal{Q})$$

Hint (i) ρ or $\mathcal{Q}$ is non invertible

(ii) ρ and $\mathcal{Q}$ are invertible

$f(B) := \det\begin{bmatrix} B & 0 \\ 0 & \mathcal{Q} \end{bmatrix} \Big/ \det(\mathcal{Q})$

$f(B)$ satisfies 1.11 and $\exists$ a det map.

* Note 1.16 — Discussions from 1.1 to 1.15 can be extended to complex matrices.

* Defn 1.17 — S_n: Symmetric group: Group of permutations of n objects

Cyclic notation for elements of S_n

* Defn 1.18 — GL_n: General linear group

$$GL_n := \{ A \in \mathbb{R}^{n\times n} \text{ s.t } AB = I \text{ where } B \in \mathbb{R}^{n\times n} \}$$

* Defn 1.19 SL_n : Special linear group

$$SL_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t } \det(A) = 1 \}$$

* Defn 1.20 O_n : Orthonormal group

$$O_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t } A \cdot A^T = A \cdot A = I \}$$

* Defn 1.21 SO_n : Special Orthonormal group

$$SO_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t } A \cdot A^T = I \text{ and } \det(A) = 1 \}$$

* Lemma 1.22 Let A and B are matrices and x a column vector s.t.

(i) $Ax = 0$

(ii) $BAx = 0$

(a) $\text{rank}(B) = \#\text{columns of } B \implies \dim(\text{solution space of (i)}) = \dim(\text{solution space of (ii)})$

(b) $\text{rank}(B) < \#\text{columns of } B \implies \dim(\text{solution space of (i)}) \leq \dim(\text{solution space of (ii)})$

Groups

* **Defn 2.1** Law of composition on a set S is a function $f : \underbrace{S \times S}_{\text{cartesian product}} \to S$

* **Def 2.2** $e \in S$ is called identity w.r.t a law of composition on set S if $\forall a \in S \quad e.a = a.e$.
Caution: A set might not contain an identity element w.r.t a law of composition

* **Defn 2.3** A transposition is an element of S_n that switches only two objects.

* **Defn 2.4** Group: A set G together with a law of composition $(.)$ s.t
① $(a.b).c = a.(b.c) \quad \forall a, b, c \in G$
② $\exists e \in G$ s.t $ea = a.e \quad \forall a \in G$
③ $a \in G \Rightarrow \exists a^{-1} \in G$ s.t $a.a^{-1} = a^{-1}.a = e$

* **Example 2.5** S_3 : Symmetry group of 3 elements
$x = (123) \quad y = (12)$ generators
$x^3 = 1 \quad y^2 = 1 \quad yx = x^2y$ defining relations
$S_3 = \{1, x, x^2, y, xy, x^2y\}$

* **Defn 2.6** Subgroup: A non empty subset $H \subset G$ s.t
① if $a, b \in H$ and $c = a.b \Rightarrow c \in H$
② if $a \in H \Rightarrow a^{-1} \in H$

* **Example 2.7** $SO_n \subset SL_n \subset GL_n$
C^∞

* **Defn 2.8** $H \subset G$ is called normal subgroup, ie $H \trianglelefteq G$
if $\forall g \in G$ and $h \in H \quad ghg^{-1} \in H$

* **Defn 2.9** Order of a group G is the number of unique elements it has.

* Defn 2.10 Let U be the subset of group G. Then the group generated by U is defined as the smallest subgroup H of G that contains all elements of U

* Defn 2.11 Cyclic subgroup: The group generated by a single element x of a group

$$\langle x \rangle := \{ \ldots, x^{-2}, x^{-1}, e, x, x^2 \ldots \}$$

Order of $x :=$ Order of $\langle x \rangle$

* Defn 2.12 Group Homomorphism: $\varphi : G \longrightarrow G'$ s.t

$$\varphi(a.b) = \varphi(a).\varphi(b) \quad \forall\, a, b \in G$$

Image of $\varphi := \{ \varphi(g) : g \in G \}$

Kernel of $\varphi := \{ g \in G : \varphi(g) = e' \}$

* Lemma 2.13 (i) $\varphi(e) = e'$

(ii) $\varphi(a^{-1}) = \varphi(a)^{-1}$

(iii) if $H < G \implies \varphi(H) < G'$

* Lemma 2.14 (i) Kernel is a normal subgroup of G

(ii) Image is a subgroup of G'

* Lemma 2.15 A homomorphism φ is injective i.e one to one if Kernel of φ if $\{e\}$

* Defn 2.16 Isomorphism: A bijective group homomorphism

* Defn 2.17 Automorphism: An isomorphism from group to itself

* Defn 2.18 $a, b \in G$ are called conjugates if $\exists\, g \in G$ s.t $b = gag^{-1}$

* Defn 2.19 Let $H \subset G$ and $a \in G$ then

$$\text{Left coset of } a := aH = \{a.h \text{ where } h \in H\}$$

↳ All the left cosets wrt a subgroup have the same size

* Theorem 2.20 Let $H \subset G$ then $[G] = [H][G:H]$

order of G | Order of H | # of left cosets

Hint: $a \underset{\text{related}}{R} b$ if $\exists h \in H$ s.t $a = b.h$ is an equivalence relation ⇒ Left cosets are non intersecting

$[G] = [H][G:H]$ ⇐ 2.19

* Corollary 2.21 Let $\varphi : G \to G'$ be a group homomorphism,

$[G] = [\text{Kernel}]\,[\text{Image}]$

* Correspondence Theorem 2.22: Let $\varphi : G \to G'$ be a surjective group homomorphism

(i) $[H \subset G \text{ s.t } K \subset H] \xrightarrow[\text{bijective map}]{f} [\text{Subgroups of } G']$

(ii) If $\{H \lhd G \text{ and } K \subset H\} \xrightarrow[\text{bijective map}]{g} \{H' \lhd G\}$

* Hint: (a) We need $K \subset H$ (i) and (ii) for f and g to be injective

(b) we need φ to be surjective for (I) f, g to be onto

(II) $\varphi(H) \lhd G'$ where $H \lhd G$ and $K \subset H$

* Defn 2.23 Quotient Group: Let $H \lhd G$, $\bar{a} := aH$ left coset of $a \in G$ then

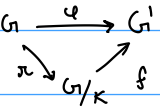
$G/H := \{\bar{a} : a \in G\}$ together with the law of composition $\bar{a}.\bar{b} = \overline{a.b}$ form a group

Further $\varphi : G \to G/H$ where $\varphi(a) = \bar{a}$ is a group homomorphism

Hint: Prove that the law of composition is independent of the choice of representative element and therefore valid. Then this law carries over the group structure.

* First Isomorphism Theorem 2.24

Let $\varphi: G \to G'$ be a surjective homomorphism and K be its kernel, then $f: G/K \to G'$ defined by $f(\bar{a}) := \varphi(a)$ is a group isomorphism.



* Lemma 2.25

If W_1 and W_2 are subgroups of G, then the following are also subgroups

$$W_1 \cap W_2 := \{ g \in G : g \in W_1 \text{ and } g \in W_2 \}$$

$$W_1 + W_2 := \{ g \in G : g = a.b \text{ or } g = b.a \text{ where } a, b \in W_1 \cup W_2 \}$$

* Usually $W_1 + W_2 \neq W_1 \cup W_2$

* Lemma 2.26

All the subgroups of $\mathbb{Z}^+$ (additive group of integers) have the form $n\mathbb{Z}$ where $n \in \mathbb{Z}$.

Hint: Use well ordering property of $\mathbb{Z}$

* GCD Defn 2.27

If $a, b \in \mathbb{Z}$, then $d \in \mathbb{Z}$ is called gcd of a and b if

(i) $d > 0$

(ii) d divides a and b

(iii) if $c \in \mathbb{Z}$ divides a and b then c divides d

* Lemma 2.28 If $a, b, d \in \mathbb{Z}$ then $d = g.c.d(a, b) \implies d\mathbb{Z} = a\mathbb{Z} + b\mathbb{Z}$

* Defn 2.29 $a, b \in \mathbb{Z}$ are called relatively prime if $g.c.d(a, b) = 1$

* Defn 2.30 Let $p \in \mathbb{Z}$ s.t if p divides $a.b \implies p$ divides either a or b

* Lemma 2.31 $p \in \mathbb{Z}$ is prime $\iff \nexists\ n \in \mathbb{Z}$ s.t $1 < n < p$ and n divides p.

* L.C.M

defn 2.32 If $a, b \in \mathbb{Z}$, then $l \in \mathbb{Z}$ is called l.c.m of a and b if

(i) $l > 0$

(ii) a and b divides l

(iii) if a and b divides $c \in \mathbb{Z}$ then l divides c

* Lemma 2.33 If $a, b, l \in \mathbb{Z}$ then $l = l.c.m(a,b) \iff l\mathbb{Z} = a\mathbb{Z} \cap b\mathbb{Z}$

* Lemma 2.34 $g.c.d(a,b) \times l.c.m(a,b) = a.b$

Proof (i) $a.b/d$ is divisible by l using 2.32

(ii) $l.d$ is divisible by $a.b$ using $d = ma + nb$

* Defn 2.48 Modular arithmetic $f: \mathbb{Z}^+ \to \mathbb{Z}/n\mathbb{Z}$ where $n \in \mathbb{Z}$ is a group homomorphism (using 2.23)

further

(i) $f(a+b) = f(a) + f(b)$

(ii) $f(a.b) = f(a).f(b)$

(iii) $f(a)(f(b) + f(c)) = f(a)f(b) + f(a)f(c)$

Congruence Relation $a \equiv b$ if $a - b$ is divisible by n

Vector Space

* Defn 3.1 A field F is a set together with two laws of composition, ie addition $(+)$ and multiplication $(\cdot)$ s.t

(i) $(F,+)$ is an abelian group with identity element denoted by 0

(ii) $(F^*,\cdot)$ is an abelian group, where $F^* = \{F - \{0\}\}$. It's identity element is denoted by 1.

(iii) $\forall\ a,b,c \in F \qquad a.(b+c) = ab + ac$

* lemma 3.2 $F_p = \mathbb{Z}/p\mathbb{Z}$ where $p \in \mathbb{Z}$ is prime is an field.

Hint: Proof by contradiction. Let $p = a.b$ with $1 < a,b < n$

show that closure fails for F_p^*

* lemma 3.3 F_p^* is an group of order $p-1$ and is cyclic, ie

$$F_p^* = \{1, \alpha, \alpha^2 \ldots, \alpha^{p-2}\} = \{\bar{1}, \bar{2}, \ldots, \overline{p-1}\}$$

The generator α of F_p^* is called primitive root modulo p.

* Defn 3.4 The characteristic of a field is the order of $\langle 1 \rangle \subset (F,+)$

* lemma 3.5

characteristic of a field

- Finite → It is a prime number
- Infinite → Then it is called zero

Hint: Same as 3.2

* Defn 3.6 A vectorspace V over a field F is a set together two laws of composition, ie. vector addition and scalar multiplication s.t

(i) V is an abelian group with scalar addition. It's identity element is denoted by 0_v

(ii) if $a \in F$, $v \in V$ then $a.v \in V$

(iii) if $v \in V$ $\quad 1.v = v$

(iv) if $v, w \in V$ and $a, b \in F \longrightarrow$

$$a(v+w) = av + aw$$

$$(a+b)v = av + bv$$

$$a.(b.v) = (a.b)v$$

* Lemma 3.7 Let $v \in V$, $0_F \in F \implies 0_F.v = 0_V$

* Defn 3.8 A subspace of a vectorspace V over field F is a non empty subset s.t

① if $v, w \in V$ then $v + w \in V$

② if $a \in F$, $v \in V$ then $a.v \in V$

* Lemma 3.9 If W_1 and W_2 are subspaces of V then

① $W_1 \cap W_2$ is a subspace of V

② $W_1 + W_2$ is a subspace of V, where $W_1 + W_2$ is the group generated by $W_1 \cup W_2$ (2.25)

* Defn 3.10 Ordered Set → Order of elements matter

→ Repetition of elements is not allowed.

{Note: It is different from real analysis}

* Defn 3.11 The Span of an ordered set $S = \{v_1, v_2 \ldots\}$ with $v_i \in V$ is defined as the subspace generated by the linear combinations of v_i's, i.e.

$$\text{Span}(S) := \left\{ \sum_i a_i v_i \text{ where } a_i \in F, v_i \in V \right\}$$

* Defn 3.12 A vector space V is called finite dimensional if $\exists$ a finite set S s.t

$$V = \text{Span}(S)$$

* Defn 3.13 An ordered set $S = \{v_1, v_2 \ldots, v_n\}$ with $v_i \in V$ is called linearly independent if

$$a_1 v_1 + a_2 v_2 \ldots + a_n v_n = 0_V \implies a_i = 0 \text{ where } 1 \le i \le n$$

* Defn 3.14 An empty set ϕ is linearly independent

$$\text{Span}(\phi) = \{0_V\}$$

* Defn 3.15 A basis $[B]$ of vectorspace V is an subset of the vectors of V s.t

① $[B]$ is linearly independent

② $\text{Span}(B) = V$

* Lemma 3.16 Let L be an linearly independent set of finite dimensional V, S be an set of V s.t $Span(S)=V$ then $\dim(Span(L)) \leq \dim(Span(S))$

Hint: Proof by contradiction $\dim(Span(L)) > \dim((Span S))$

* Defn 3.17 The dimension of a finite dimensional vectorspace V is defined as the number of vectors in its basis

* Lemma 3.18 Let V be a finite dimensional vectorspace and, $[\mathcal{B}]$ and $[\mathcal{B}]'$ are its two basis associated by $[\mathcal{B}]' = [\mathcal{B}]P \Longrightarrow$ P is invertible

P: base change matrix

If $v \in V$ then $v = [\mathcal{B}]'x' = [\mathcal{B}]x$ (x: coordinate vector of v w.r.t $[\mathcal{B}]$)

$x = Px'$

* Defn 3.19 The two subspaces U_1 and U_2 of V are called linearly independent if

$$U_1 \cap U_2 = \{0_v\}$$

* Defn 3.20 A vectorspace V is called direct sum of subspace $U_1, U_2 \dots, U_k$ if

① $U_1, U_2 \dots, U_k$ are linearly independent

② $V = U_1 + U_2 + \dots U_k$

It is denoted by $V = U_1 \oplus U_2 \oplus \dots \oplus U_k$

* Lemma 3.21 If W_1 and W_2 are subspace of V then

$$\dim(W_1) + \dim(W_2) = \dim(W_1 + W_2) + \dim(W_1 \cap W_2)$$

Linear Transformation

* **Defn 4.1** Let V, W be vectorspaces over the same field F. Then a map $T: V \to W$ is called a linear transformation if

  ① $T(v_1 + v_2) = T(v_1) + T(v_2)$ $\quad \forall v_1, v_2 \in V$

  ② $T(av) = a\,T(v)$ $\quad \forall v \in V, \forall a \in F$

* **Defn 4.2**

  Kernel of $T := \{ v \in V : T(v) = 0_W \}$

  Image of $T := \{ T(v) : v \in V \}$

* **Lemma 4.3**

  Kernel is a subspace of V

  Image is a subspace of W

* **Lemma 4.4**

  $\dim(\text{Kernel}) + \dim(\text{Image}) = \dim(V)$

  nullity $\quad$ rank

  Hint: $[B] := \{ \underbrace{a_1, a_2, \ldots, a_k}_{\text{Basis of Kernel}}, \underbrace{b_1, \ldots, b_m}_{\text{Basis of } V} \}$

  Show $\{T(b_1), T(b_2), \ldots, T(b_m)\}$ is a basis of Image

* **Lemma 4.5**

  Let $T: V \to W$ be a linear transformation, P a subspace of V

  then $T(P)$ is an subspace of the Image

* **Defn 4.6** Let $T: V \to W$ be a linear transformation, $[B], [C]$ be the basis of V and W respectively

  Then the matrix of linear transformation T wrt $[B]$ and $[C]$ is defined as a matrix A s.t

  $$T([B]) = [C]\,A$$

  Lemma If x, y be the coordinate vectors of $v \in V$ and $T(v) \in W$ wrt $[B]$ and $[C]$ respectively

  then $Y = Ax$

  Lemma If $[B']$ and $[C']$ are another sets of basis of V and W s.t

  $$[B'] = [B]\,P \qquad [C'] = [C]\,Q$$

  then A', the matrix of linear transformation T wrt $[B']$ and $[C']$ is given by

  $$A' = Q^{-1} A P$$

b/w two vectorspaces V and W over the same field F is a

* Defn 4.7 A linear transformation $T: V \longrightarrow W$ is called an Isomorphism if T is bijective.

Example: A n dim vectorspace V over field F is isomorphic to the vectorspace F^n.

* Lemma 4.8 Let $A_{n\times n}$ be an square matrix then

$$Ax = 0 \iff A \text{ is invertible}$$

if and only if $x = 0$

Proof (i) A is invertible $\Rightarrow$ Let $Ax = 0 \Rightarrow A^{-1}Ax = A^{-1}0 \Rightarrow x = 0$

(ii) Let $Ax = 0$ only for $x = 0 \Rightarrow$ columns of A are linearly independent

[A] is an basis of $\mathbb{R}^n$, [I] is another basis of $\mathbb{R}^n \therefore \exists P$ s.t $[A]\underbrace{P}_{\text{invertible}} = [I]$

* Lemma 4.9 Let $A_{m\times n}$ be a matrix and $E_{m\times m}$ be an invertible matrix, then

$$\dim(\text{colSpace}(A)) = \dim(\text{Column Space}(EA))$$

Hint use 4.4, 4.5

* Lemma 4.10 Let $A_{m\times n}$ be a matrix then

$$\dim(\text{colSpace}(A)) = \dim(\text{RowSpace}(A))$$

Hint Use 1.8, 4.9

* Defn 4.11 A linear operator T is an linear transformation from a vectorspace V to itself.

- Invertible / T is bijective / kernel $= \{0_v\}$ / Image $= V$ → Transformation matrix is invertible
- Singular → Transformation matrix is non invertible

* Lemma 4.12 Let $T: V \to V$ be a linear operator, then the following conditions are equivalent

(i) $V = \text{kernel}(T) \oplus \text{Image}(T)$

(ii) $\text{kernel}(T) \cap \text{Image}(T) = \{0_v\}$

(iii) $\text{kernel}(T) + \text{Image}(T) = V$

These don't mean T is invertible

Eg: $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Hint: Use 4.4, 3.21

* Lemma 4.13 Let $T:V \longrightarrow V$ be a linear operator, $[B], [B']$ be two different basis of V associated by $[B'] = [B]P$ and A, A' be the transformation matrix wrt $[B]$ and $[B']$ then

$$A' = P^{-1}AP, \text{ further}$$

A and A' are called similar matrices as they represent the same linear transformation

* Defn 4.14 Let $T:V \longrightarrow V$ be a linear operator, then a subspace W of V is called T invariant if $T(W) \subset W$

* Lemma 4.15 Let $T:V \longrightarrow V$ be an linear operator, W be an T-invariant subspace of V, $\{w_1, w_2 \ldots, w_k\}$ be an basis of W, $[B] := \{w_1, w_2 \ldots, w_k, v_1, v_2 \ldots, v_n\}$ be a basis of V, $f: V \longrightarrow V/W$ be a map defined by $f(v+W) := \bar{v}$

(i) $\bar{v}_1 + \bar{v}_2 = \overline{v_1 + v_2}$

(ii) $a\bar{v}_1 = \overline{av_1}$

} Similar to 2.23

(iii) V/W along with (i) and (ii) forms a vectorspace

(iv) $f: V \longrightarrow V/W$ is a linear transformation

(v) $[\bar{B}] := [\bar{v}_1, \bar{v}_2 \ldots, \bar{v}_n]$ is an basis of V/W.

(vi) $\varphi: V/W \longrightarrow V/W$ given by $\varphi(\bar{v}) := f(T(v))$ is an linear operator

(vii) The transformation matrix of T wrt $[B]$ has the form $A = \begin{bmatrix} P_{k\times k} & Q_{k\times n} \\ 0_{n\times k} & R_{n\times n} \end{bmatrix}$

(vii)(a) The transformation matrix of φ wrt $[\bar{B}]$ is $R_{n\times n}$

* Lemma 4.16 An eigen vector v of a linear transformation $T:V \longrightarrow V$, where V is a vectorspace over field F, is an (i) non zero vector

(ii) $T(v) = \lambda v$ where $\lambda \in F$

* Note 4.17 An eigen vector is an one dimensional T-invariant subspace

* Lemma 4.18 Let V be a vectorspace over field F, $T:V \longrightarrow V$ an linear operator, $\lambda \in F$ then $\lambda I - F: V \longrightarrow V$ is an operator

* Lemma 4.19 Let $v \in V$ be an eigen vector of $T: V \to V$ with eigen value λ, then

characteristic polynomial of T $\{ \det(\lambda I - A) = 0$ where A is the matrix of T $\Longleftarrow$ $\lambda I - T$ is singular $\Longleftarrow$ $v \neq 0$ is in the null space of $\lambda I - T$

* Lemma 4.20
(i) The characteristic polynomial of $T: V \to V$ is independent of the choice of basis
(ii) Similar matrices have the same characteristic polynomial
(iii) The co-efficients of characteristic polynomial is independent of the choice of basis
(iv) Eigen values are roots of the characteristic polynomial

* Lemma 4.21 Eigen vectors corresponding to distinct eigen values are linearly independent

Hint: Proof by induction.

* Lemma 4.22 If $T: V \xrightarrow{n\text{ dim}} V$ has n distinct eigen values the $\exists$ basis $[\mathcal{B}]$ of V s.t the transformation matrix of T is diagonal, ie. $\begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_n \end{bmatrix}$

* Schur's Lemma 4.23 Let $T: V \xrightarrow{n\text{ dim v.s over } F} V$ be a linear operator,

If the characteristic polynomial of T factors into linear terms $\Longrightarrow$ $\exists$ basis $[\mathcal{B}]$ of V s.t the transformation matrix of T is upper diagonal $\begin{bmatrix} R_1 & & * \\ & \ddots & \\ 0 & & R_n \end{bmatrix}$

Hint Use 4.20(iv), 4.17, 4.15 (vi), (viii) to prove upper diagonal

Proof it is not always diagonalizable :- Let v_1 be the eigen vector of $T: V \to V$, V_1 the span of v_1, $f: V \to V/V_1$ where $f(v_2 \in V_1) = \bar{v}$

Let $\bar{v}_2$ be the eigen vector of $\varphi: V/V_1 \to V/V_1$ $\varphi(\bar{v}) := f(T(v))$

$v_2' = v_2 + c v_1$ $\quad T(v_2) = \lambda_2 v_2 + n v_1$

$T(v_2') = T(v_2) + T(c v_1) = \lambda_2(v_2' - c v_1) + c\lambda_1 v_1 = \lambda_2 v_2' + (n + c(\lambda_1 - \lambda_2)) v_1$

then if $\lambda_1 \neq \lambda_2$ we can diagonalize

if $n = 0$ we can diagonalize

if $\lambda_1 = \lambda_2$ and $n \neq 0$ we cannot diagonalize

* Defn 4.24 A generalized eigen vector x, with eigen value λ and exponent d of $T: V \to V$ is a vector s.t

① $x \neq 0_v$

② $(T-\lambda I)^d x = 0$

③ If $(T-\lambda I)^k x = 0 \implies k \geq d$

* Lemma 4.26 Let $x \in V$ be a generalized eigen vector x of $T: V \to V$ with eigen value λ and exponent d,

$$u_i := (T-\lambda I)^i x \quad \text{with} \quad 0 \leq i \leq d-1$$

$$[\mathcal{B}] := \{u_1, u_2 \ldots, u_{d-1}\}$$

$$\mathcal{X} := \text{Span}([\mathcal{B}])$$

then

(i) $\mathcal{B}$ is linearly independent

(ii) $\mathcal{X}$ is T invariant

(iii) $T([\mathcal{B}])$ has the form $\begin{bmatrix} \lambda & & 0 \\ 1 & \lambda & \\ & 1 & \lambda \\ 0 & & 1 & \lambda \end{bmatrix}_{d \times d}$

Proof Hint (i) Show that any non trivial combination $[\mathcal{B}]$ is an generenerelized eigenvector $\therefore \neq 0_v$

* Lemma 4.25 A generalized eigen vector x of $T: V \to V$ has unique eigen value λ and exponent d

Proof Hint: Use binomial expansion

$$(x+y)^n = \sum_{k=0}^{n} {}^nC_k \, x^k y^{n-k}$$

* Jordan Decomposition Theorem 4.26 Let T be a linear operator on a finite dimensional complex vectorspace V. Then $\exists$ a basis $[\mathcal{B}]$ of V s.t the transformation matrix of T w.r.t $[\mathcal{B}]$ has the Jordan form.

Proof: We will prove as they together lead to the basis $[\mathcal{B}]$ wrt which the transformation matrix of T has the jordan form.

① $V = V_1 \oplus V_2 \oplus \cdots \oplus V_r$

② $V_i = \text{Span}([u_{i,0}, u_{i,1} \ldots, u_{i,d_i-1}])$, $1 \leq i \leq r$

③ $u_{i,j} = (T-\lambda_i) u_{i,j-1}$, $1 \leq j \leq d_i - 1$

generalized eigen vector with eigen value λ_i and index d_i

We know that ①, ②, ③ are true for 1 dim V.

We will use principle of induction, ie assuming the statement is true for complex vector space of dimensions 1 to n-1 prove it for n dim complex vectorspace.

* Let $T: V \to V$ be a linear operator on n dim complex vector space V. Since the characteristic polynomial factors into linear terms in complex field $\exists$ an eigen vector v with eigen value λ.

If w.r.t basis [B] T has jordan form then so does $T-\lambda I$. Therefore for simplicity we we work with $T+\lambda I$ and refer to it as T, then

$T: V \longrightarrow V$ is an linear operator and its nullspace is non-trivial.

Let K_i be the kernel and U_i be the image of $T^i: V \to V$. Since V is finite dimensional let $m \in \mathbb{N}$ be the smallest number s.t

$$K_i = K_j \quad \text{and} \quad U_i = U_j \qquad \forall\ i, j \geq m$$

then let $v \in K_m \cap U_m$, since $T^m v = 0$ and $\exists\ x \in V$ s.t $T^m x = v$

$T^{2m} x = 0 \Rightarrow x \in K^m \quad \therefore \quad v = T^m x = 0$

Using 4.12 $V = K_m \oplus V_m$

$\dim(V_m) < n$ $\therefore$ $\exists$ jordan generators. If $\dim(K_m) < n$ the proof is complete otherwise T is an idempotent operator

* $T: V \to V$ be an idempotent eigen vector, prove $\exists$ a basis [B] s.t T has jordan form.

Let K be the kernel of T and U be the image of T. Since $\dim(U) < n$ $\exists$ jordan generators for it ie. $U = U_1 \oplus U_2 \cdots \oplus U_r$ and

$$U_i = \text{Span}([u_{i,0}, u_{i,1} \cdots u_{i,d_i-1}]), \qquad 1 \leq i \leq r,$$

$$u_{i,j} = (T)^j v_i, \qquad 1 \leq j \leq d_i - 1$$

generalized eigen vector with eigen value 0 and index d_i

Let $w_i \in V$ s.t $T w_i = v_i$ then w_i is an generalized eigenvector with eigen value 0 and index $d_i + 1$

Let $W_i = \text{Span}([u_{i,0}, u_{i,1} \cdots u_{i,d_i}])$ where $u_{i,j} = T^j w_i$

$$W = W_1 + W_2 + \cdots + W_r$$

let $w \in W_n \cap W_k$ then $Tw \in V_n \cap V_k$ since $V_n \cap V_k = 0$

$w \in K$ Since $W_n \cap K = \text{Span}(u_{n,d_n}) = \text{Span}(v_{n,d_n-1})$

and $W_k \cap K = \text{Span}(u_{k,d_k}) = \text{Span}(v_{k,d_k-1})$

since v_{n,d_n-1} and v_{k,d_k-1} are linearly independent

$w = 0_v$

$\therefore W = W_1 \oplus W_2 \oplus \cdots \oplus W_r$

let P = subspace of K that is not in W

Augment basis of P to W to get $[B]$ s.t T has jordan form w.r.t it.

* lemma 4.27 If T is a linear operator on a finite dimensional complex vectorspace V.

If for some $n \in \mathbb{Z}$, T is the identity operator ie $T^n = I \Rightarrow T$ is diagonalizable

Orthogonal Linear Operator

* Defn 5.1 $A \in \mathbb{R}^{n \times n}$ is called orthogonal if $A^T A = I$

* Dot product 5.2 Let $u, v \in \mathbb{R}^n$, $\langle u, v \rangle := x_1 y_1 + x_2 y_2 - - + x_n y_n$ where $X = \{x_1, x_2 - -, x_n\}^t$, $Y = \{y_1, y_2 - -, y_n\}^t$ are the coordinate vectors of u, v wrt the standard basis

* Defn 5.3 Let $v \in \mathbb{R}^n$, then $\underbrace{\|v\|}_{\text{length of vector}} := \langle v, v \rangle$

* Defn 5.4